

HW 1 (Due 9.23)

(10 pts each, 100 pts in total.)

1. Prove that $\mathbb{R}$ is uncountable.
2. ((iii) $\Rightarrow$ (i) in Heine–Borel theorem) Prove that for $K \subseteq \mathbb{R}^n$, if every sequence in K has a subsequence converging to a point of K , then K is compact.
3. Prove that continuous maps preserve connectedness.
4. Prove that an open disc in $\mathbb{R}^2$ is not a disjoint union of open rectangles.
5. Prove the following proposition: Suppose

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = A,$$

and suppose that for every $x \neq x_0$ sufficiently near x_0 the limit

$$\phi(x) = \lim_{y \rightarrow y_0} f(x,y)$$

exists. Then $\lim_{x \rightarrow x_0} \phi(x) = A$.

6. Set

$$f(x,y) = \begin{cases} \frac{x^2 y}{x^4 + y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Is f separately continuous at $(0,0)$? Is f (jointly) continuous at $(0,0)$?

7. Write $R = [a,b] \times [c,d]$. Prove the following theorem: Suppose $f : R \rightarrow \mathbb{R}$ is continuous and

$$F(t) := \int_a^b f(x,t) dx.$$

For each fixed $x \in [a,b]$, regard

$$g_x(t) = f(x,t)$$

as an ordinary one-variable function of t . Suppose every g_x is differentiable on (c,d) and that the function

$$k(x,t) = g'_x(t)$$

is continuous on R . Then F is differentiable on (c, d) and

$$F'(t) = \int_a^b k(x, t) dx.$$

8. Let f be a strictly positive function continuous on $[0, 1]$. Find the discontinuity point of

$$I(y) = \int_0^1 \frac{yf(x)}{x^2 + y^2} dx \quad (y \in \mathbb{R}).$$

9. The interior of a subset S of a topological space X is the union of all subsets of S that are open in X . Are interiors of connected sets in $X = \mathbb{R}^2$ always connected? What if $X = \mathbb{R}$? Explain why.
10. A subset S of a topological space X is said to be dense in X provided that the closure of S in X is equal to X . Prove the Baire Category Theorem: If $\{U_n\}_{n=1}^\infty$ is a sequence of open dense subsets of $\mathbb{R}^n$, then $\bigcap_{n=1}^\infty U_n$ is dense in $\mathbb{R}^n$. Do we have the same conclusion if we replace $\mathbb{R}^n$ by the unit open ball $B_1(0)$? Explain why.